

Finance & Government

Required Materials

25 Mar 2026

Bahoo, S., Cucculelli, M., Goga, X., & Mondolo, J. (2024). Artificial intelligence in Finance: A comprehensive review through bibliometric and content analysis. *SN Business & Economics*, 4(2), 23. <https://doi.org/10.1007/s43546-023-00618-x>

Mellouli, S., Janssen, M., & Ojo, A. (2024). Introduction to the Issue on Artificial Intelligence in the Public Sector: Risks and Benefits of AI for Governments. *Digital Government: Research and Practice*, 5(1), 1–6. <https://doi.org/10.1145/3636550>

Artificial intelligence in Finance

Main Idea

- this paper argues that **AI** has become a **major force in finance research and practice**
- the literature **grows** sharply **after 2015**
- the **biggest** application **areas** are:
 - **prediction / forecasting**
 - **classification / detection / early warning**
 - big **data**, text **mining**, and sentiment **analysis**

What the Authors Found

- the **literature** covers **74 countries**
- the **most studied** regions
 - **Europe**
 - **United States**
 - **China**
- **banking** and **financial** services **dominate**, but the **research** also **spans** other **sectors**
- most studies are **quantitative**
- common **methods**
 - **support vector machines**

	○ random forests • advanced neural nets such as LSTM and HONN often perform well, but they are harder to implement than simpler models
Three Biggest Use Cases	• forecasting stock prices, performance, and volatility: 39 papers • classification and warning systems for fraud, credit risk, and firm/bank monitoring: 23 papers • text mining, NLP, sentiment analysis: 14 papers
Top 10 Research Streams	stock market • predict stock prices • study algorithmic trading • the paper claims ○ AI models can capture nonlinear relationships better than linear models ○ algorithmic trading can improve liquidity and efficiency ○ high-frequency trading can also amplify some volatility trading models • researchers use neural networks and machine learning to build automated trading systems that choose positions and manage risk • these systems can generate profitable trades under controlled risk in some setting volatility forecasting • forecasting volatility is difficult • classic models like HAR often remain very strong • newer neural-network approaches ○ LSTM (Long Short-Term Memory)

	▪ a specialized Recurrent Neural Network (RNN) architecture designed to learn long-term dependencies in sequential data, overcoming the vanishing gradient problem of traditional RNNs ○ NARX (Nonlinear Autoregressive Exogenous) ▪ a specialized type of recurrent neural network (RNN) used for time series modeling and system identification ○ HONN (Higher-Order Neural Networks) ▪ a type of artificial neural network where the input to a computational neuron is a weighted sum of products of its inputs, rather than just a simple weighted sum • claim: ○ in some cases, HONN outperforms more standard neural net
Top 10 Research Streams (cont.)	portfolio management • choose assets • estimate risk • improve index tracking • optimize allocation • claim: ○ AI-based prediction can improve portfolio selection by giving better return forecasts and more refined risk handling

performance, risk, and default valuation

- predict default
 - firm distress
 - bankruptcies
 - mortgage and loan defaults
 - bank performance
- claim:
 - AI-powered classifiers often outperform classic linear models in these task

cryptocurrencies

- this stream is newer and less mature
- Bitcoin return modeling
- volatility analysis
- arbitrage
- risk management
- claims:
 - area is still developing
 - human traders still dominant
 - regulation still immature

derivatives

- option and derivative pricing
- claims
 - some neural-network methods outperform traditional pricing approaches
 - the review also notes disagreement
 - not every deep-learning approach is stable

- some hybrid methods may work better

credit risk in banks

- bank failure prediction
- credit default prediction
- fraud detection
- early-warning systems
- credit scoring
- claims:
 - arguably the most practically important areas
 - strong predictive performance, including cases where credit events can be predicted well in advance
 - transparency remains a concern

investor sentiment analysis

- AI uses text from news and social media to estimate investor sentiment
- that sentiment can then be used as an input for
 - price direction
 - liquidity
 - return prediction
- claim:
 - this as an important bridge between NLP and finance

foreign exchange management

- forecast exchange-rate movement
- support trading or hedging decisions
- claim:

	○ several AI models outperform older time-series or standard machine-learning approaches in forex prediction
Limits and Open Problems	• the paper is not claiming AI has solved finance • the authors highlight several unresolved issues ○ models need more testing during crisis periods and black swan events ○ some methods are still black boxes ○ overfitting remains a concern ○ more work is needed on systemic-risk monitoring ○ some areas need richer inputs, such as ▪ order-book data ▪ market data ▪ social media ▪ text-based signals ○ crypto research needs more attention to regulation and blockchain development ○ derivative-pricing models still need work on computational efficiency and better reasoning design
Introduction to the Issue on Artificial Intelligence in the Public Sector	
Main Point	• AI is becoming an important part of government work around the world • governments are using AI to ○ improve services ○ support internal operations ○ strengthen digital transformation ○ creates social, ethical, legal, and governance problems

	• claim: ○ so it cannot be treated as a purely technical tool
The Scope of 'AI' Chosen by the Authors	• machine learning/deep learning • natural language processing • robotics • computer vision • generative AI
What the Authors are Asking	claim: • governments are interested in AI in two main ways ○ How can AI improve government services and operations? ○ How can governments reduce AI's risks and make sure it serves the public good?
Benefits the Paper Highlights	claim: • AI can help governments by ○ improving service quality ○ increasing productivity ○ reducing costs ○ supporting decision-making ○ automating repetitive tasks ○ extracting information from large amounts of data ○ making faster responses possible ○ enabling more continuous service availability ○ helping with complex problems and evidence-based decisions • the paper sees AI as a tool that can make government

	○ faster ○ more data-driven ○ more operationally capable
Risks the Paper Highlights	● AI can create serious problems ○ job displacement ○ loss of human expertise ○ high costs and energy use ○ misinformation and fake content ○ less accountability and less transparency ○ biased training data ○ biased or unfair decisions ○ systemic discrimination ○ privacy invasion ○ non-explainable decisions ○ reduced public trust ○ weak governance of data and technology ● claim: ○ AI may help governments act faster, but it can also make them less fair, less understandable, and harder to hold accountable if it is poorly governed
What the Special Issue Covers	● emergency response ● policy analysis ● fraud detection in public bids ● citizen participation and public consultation ● legal transparency and freedom-of-information questions ● bureaucratic decision-making

	• AI capability and governance • algorithmic government and web-material analysis
What Each Contribution Basically Says	commentary on the judicial system • one piece looks at AI in the Italian judicial system which identifies uses such as ○ legal-document analysis ○ process modeling ○ real-time monitoring ○ predictive techniques ○ workload analysis • it also argues for explainability and domain-adapted language models transparency and freedom of information • another article studies whether freedom-of-information rules can make government AI transparent • its conclusion is cautious ○ FOI often gives access only to existing documents ○ access to algorithms or documentation may be blocked by proprietary claims AI in bureaucratic decision-making • a Weberian analysis argues that AI in bureaucracy raises problems involving ○ transparency ○ accountability ○ bias ○ systemic discrimination

- **claim:**

- these are not minor technical flaws
- they affect how power is exercised over citizens

can AI replace policy analysts?

- one article tests whether NLP systems can generate useful policy briefing notes
- **the answer**
 - not fully
 - the tools were not able to produce useful briefings on their own
 - but they may still help human analysts as a supplement

fraud detection in public bids

- a paper called PLUS proposes a semi-automated pipeline for detecting fraud or irregularities in public bidding documents
- the summary says it helps reduce specialists' workload when searching for suspicious patterns

emergency response

- another paper presents a human-centered AI tool for proactive incident detection using crowdsourced data
- the reported result is that its CNN-based method detects incidents better than baseline methods

	citizen participation • one review looks at NLP methods for analyzing citizen input in public consultation • the paper says AI can help ○ group responses ○ identify arguments ○ detect opinions • there are still important problems with ○ quality ○ non-English data ○ usable software for practitioners
	AI capability and governance • another article examines data and technology governance for AI • its conclusion is that stronger IT and data governance can create the foundation needed for AI capability in government
	algorithmic government and web materials • the final article uses text embeddings and k-means clustering to analyze web documents related to algorithmic government • the goal is to help governments understand citizen concerns more quickly and support more evidence-based decisions
Future Questions	• how AI changes the work of civil servants • how to maintain transparency and accountability • how to achieve trustworthy AI outcomes • which factors really drive AI adoption • how to gain AI's benefits without worsening its risks

- how to make AI implementation succeed in practice
- what regulation and governance frameworks are needed
- how AI will transform government over time

Podcast

Inside Anthropic's Standoff with the Pentagon and What It Means for Military AI

Center for Strategic & International Studies (CSIS) – *The AI Policy Podcast*

- Gregory C. Allen
- Matt Mande

Inside Anthropic's Standoff with the Pentagon: Overview

Purpose

- CSIS's *The AI Policy Podcast*
 - released on **February 25, 2026**
- **policy discussion** covering three broad topics
 - the Anthropic–Pentagon dispute
 - AI's economic effects
 - claims about DeepSeek using Nvidia Blackwell chips
 - frontier-model distillation
- claims:
 - the fight between Anthropic and the U.S. Department of Defense is not just a contract dispute
 - it is really a conflict over who gets to set the rules for military use of advanced AI
 - the government
 - the company that built the model
 - the conversation then widens to two related issues
 - how AI is already disrupting white-collar industries
 - how national-security competition is shaped by chips, export controls, and model distillation

Part 1: Anthropic vs. the Pentagon	
What the Conflict is About	- the Pentagon is frustrated by Anthropic's usage restrictions on Claude, especially for military applications - the official description frames this as an "escalating" clash - the discussion centers on the DoD's desire for broader use rights versus Anthropic's insistence on keeping some safeguards in place
The Government's Position	- the Pentagon's basic view - if it is buying access to a powerful model for national-security purposes, it does not want a private company imposing extra limits on what the government can do with that capability - in other words, the military wants maximum operational flexibility
Anthropic's Position	- Anthropic's side of the dispute is presented as an effort to preserve model safeguards and maintain some boundaries around military use - the speakers treat this as a question of whether an AI company can or should enforce ethical or safety limits even when the customer is the U.S. government
Why it Matters	- the podcasters treat this as a high-stakes test case for military AI governance - it is not only about one company or one contract, this is about whether future frontier-model providers will be expected to give the government unrestricted access once a lawful contract exists

the answer is:
YES!...?

The Ultimatum and The Stakes	• Defense Secretary Pete Hegseth’s ultimatum and the possible consequences of labeling Anthropic a “supply chain risk” or invoking the Defense Production Act • the podcast treats this as the most coercive part of the conflict ○ if Anthropic refuses the terms, the government may have tools to pressure or bypass the company
The Hard Question	• should frontier AI labs be able to keep safety red lines in place for military customers? • or does the government get final control once it purchases the technology for lawful national-security use?
Part 2: AI’s Economic Disruption	
The Discussion	• the second major segment shifts away from defense and looks at how AI is affecting the economy • the discussion covers ○ recent stories driving discussion about the economic impacts of AI ○ industries primarily discussed ▪ auditing ▪ software markets ▪ filmmaking
Auditing and Professional Services	• the hosts discuss AI’s impact on auditing and related white-collar work • the implication is that AI is already changing expectations about how much human labor some knowledge-work tasks really require • in professional service industries this could put pressure on

	○ pricing ○ staffing ○ margins
Software and Broader Business Effects	• the underlying idea appears to be that if AI can do more of the work that software once helped humans do, then some companies may gain value while others may lose it, depending on where automation pressure lands
Filmmaking and Creative Work	• Hollywood and filmmaking are faced with AI's possible encroachment into creative production • AI could change ○ workflows ○ labor demand ○ the meaning of authorship
The Big Picture	• the episode's broader message is that AI is not only a defense and policy story • it is also an economic restructuring story • especially for highly paid cognitive and creative work
Part 3: DeepSeek, Chips, and National Security	
Topic	• the final section turns to a national-security and export-control issue • the hosts discuss a senior government official's claim that DeepSeek's forthcoming model was trained using Nvidia Blackwell chips and frontier model distillation

Why This Matters	• if their claim is true, it would raise two major concerns ○ are advanced chips are reaching restricted actors despite export controls ○ can leading-model capabilities be replicated or accelerated through distillation from frontier systems
Broader Implication	• the speakers ask what these allegations mean for export-control effectiveness and for strategic competition • the concern is that even strong restrictions may fail if hardware can be smuggled or if model capabilities can be copied indirectly
Section Wrap-Up	• This section treats AI competition as a supply-chain and security problem, not just a software problem. Chips, access controls, and IP leakage matter as much as model design
Conclusion	
A Three-Front Issue	• the episode presents AI as a three-front issue ○ a governance fight over military use ○ an economic force reshaping professional and creative labor ○ a geopolitical technology race shaped by chips, controls, and model diffusion • the video's core argument ○ AI policy can no longer be separated into neat categories • the same frontier systems raise questions about war powers, contractor control, labor disruption, industrial strategy, and export enforcement all at once

- the Anthropic–Pentagon dispute is the entry point, but the real theme is that AI is becoming a system-level issue for both government and society